Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

Object Oriented Programming

BCA — Semester 2 — — Answer Sheet
Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. What do we need template? Why the program terminate after the occurrence of exception? Describe the types of constructor.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Differentiate between abstract class and concrete class. Write a program to create a class named "Quadratic" that represents a function of the form f(x) = ax + bx + c, where x is a real variable and a, b, c are real constants. The class must satisfy the following requirements.a. A constructor should be provided that takes the values of a, b and c as arguments. All three of these arguments should default to zero.b. A function that takes a single argument x return the value of f(x).c. All data members should be private.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain the chain of constructor and destructor between sub classes and super classes during inheritance with example. Create a class named "Point" with data members x and y. Overload the + operator to add the value of two objects of this class.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Can we use object oriented concept using structure instead of classes? Justify your opinion.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. In which case default argument is used? Describe with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is the task of static data members and function? Illustrate with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Differentiate between aggregation and inheritance. Why do we need virtual base class?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define stream. How do you read the text from file? Describe.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Explain the features of OOP.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Define namespace. List any two manipulators with their uses.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Why do we need friend class? How can we define member function outside the class?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Explain the use of pure virtual function with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.